October Batch — Practical Assessment

CYBERSECURITY FUNDAMENTALS

Topics: Networking Fundamentals & Linux Fundamentals

Assessment Type: Practical + Scenario-Based Questions

Total Marks: 100

Difficulty: Beginner to Intermediate

Submission Format: PDF with command outputs, calculations, and screenshots

SECTION 1: NETWORKING FUNDAMENTALS

Total Marks: 50

Task 1 — Case Study: Design a Network for an Office Building

Marks: 25

Scenario: A Four-Floor Corporate Office

A company is setting up a new four-floor office building. Each floor has different departments, and the IT team needs to design a network that supports current employees and future expansion.

The company has provided the following requirements.

Floor	Department	Current Devices
Ground Floor	Reception and Administration	20
First Floor	HR and Finance	35
Second Floor	Development Team	60
Third Floor	SOC and IT Operations	45
Server Room	Internal Servers and Network Services	12

Additional requirements:

- The company has received the private IP network `192.168.10.0/24`.
- Each floor should have its own subnet.
- The network should support approximately 20% additional device capacity per department.
- The server room should be separated from employee networks.
- Employees working remotely should be able to access authorized internal resources through a VPN.
- The network administrator wants to reduce unnecessary communication between departments.
- Network devices such as routers, switches, and access points also require IP addresses.

Your Mission

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You have been assigned the role of a junior network engineer. Design a network addressing plan that meets the company's requirements.

Questions

Q1. Calculate Device Requirements

Calculate the required device capacity for each department, including the additional 20% capacity.

Show your calculations.

Q2. Perform Subnetting

For each department:

1. Determine the minimum suitable subnet size.
2. Identify the subnet mask and CIDR prefix.
3. Calculate the network address.
4. Calculate the first usable IP address.
5. Calculate the last usable IP address.
6. Calculate the broadcast address.
7. Calculate the total number of usable host addresses.

Q3. Create an IP Address Allocation Table

Complete the following table.

Department	Network Address/CIDR	Subnet Mask	First Usable IP	Last Usable IP	Broadcast Address	Usable Hosts
Reception and Administration						
HR and Finance						
Development Team						
SOC and IT Operations						
Server Room						

Ensure that the subnets do not overlap and fit within the assigned network.

Q4. Network Capacity Analysis

Answer the following:

1. How many usable IP addresses are available in the original `/24` network?
2. Can all five departments fit into the assigned network while meeting the expansion requirements?
3. If the network is insufficient, what alternative addressing plan would you recommend?
4. Why should network engineers plan for future growth rather than allocate addresses only for current devices?

Q5. Network Design

Draw a simple network diagram showing:

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- Internet connection.
- Edge router or firewall.
- Core switch.
- Four departmental networks.
- Server network.
- Remote employees connecting through a VPN.

You may use a diagram tool or draw it manually and insert a clear photograph into your report.

Q6. Security Recommendations

Suggest at least four security controls for this office network.

Consider:

- Departmental network separation.
- Firewall rules.
- VPN authentication.
- Server access restrictions.
- Network monitoring.
- Secure Wi-Fi configuration.

Explain why each control is necessary.

Evidence Required

- Subnetting calculations.
- Completed IP allocation table.
- Network diagram.
- Written explanation of whether the assigned address space is sufficient.
- Security recommendations.

Task 2 — Networking Concepts and Troubleshooting

Marks: 15

Answer the following questions in your own words.

Q1. IP Addressing

Explain the difference between a private IP address and a public IP address. Give two examples of private IPv4 address ranges.

Q2. Subnetting

What is subnetting, and why do organizations divide a large network into smaller subnets?

Q3. Default Gateway

A computer has the following configuration:

- IP Address: `192.168.1.25`

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- Subnet Mask: `255.255.255.0`
- Default Gateway: `192.168.1.1`

Explain the role of the default gateway when this computer communicates with a device on another network.

Q4. DNS

A user can access a website using its IP address but cannot access it using its domain name.

What could be the problem? Mention at least two possible causes and explain how you would investigate them.

Q5. DHCP

Explain how DHCP helps devices obtain network configuration automatically.

What problems might occur if a computer cannot obtain an IP address from the DHCP server?

Q6. TCP vs. UDP

Explain the difference between TCP and UDP. Give one practical example of an application or service that uses each.

Q7. VPN

An employee is working from home and needs to access an internal company server.

Explain how a VPN can help establish secure remote access. Also mention one security risk associated with poorly configured VPN access.

Q8. Network Troubleshooting

A computer can communicate with other devices on its local network but cannot access the internet.

Describe the order in which you would investigate the problem. Explain the purpose of each check.

Task 3 — Practical Networking Commands

Marks: 10

Perform the following tasks on your computer or an authorized lab machine.

Activity 1: Identify Network Configuration

Use appropriate commands to identify:

- Your device's IP address.
- Subnet mask or CIDR prefix, where available.
- Default gateway.
- DNS servers.
- Network interface names.

Linux hint: `ip addr` and `ip route`

Windows hint: `ipconfig /all`

Activity 2: Test Network Connectivity

Use suitable commands to:

1. Check whether your local gateway is reachable.

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2. Check connectivity to an external IP address.
3. Investigate whether a domain name resolves correctly.
4. Display the route packets take toward a destination.

Possible commands include:

- `ping`
- `ip route`
- `nslookup`
- `traceroute` or `tracepath`
- `tracert` on Windows

Activity 3: Interpret Your Results

Answer the following:

1. What does a successful ping indicate?
2. Does a failed ping always mean the destination is unreachable? Explain.
3. What is the purpose of DNS lookup?
4. What information can a traceroute provide?
5. Why might a device have an IP address but still lack internet connectivity?

Evidence Required

Include screenshots of command outputs and briefly explain what each result tells you.

SECTION 2: LINUX FUNDAMENTALS

Total Marks: 50

Task 4 — Linux Filesystem Navigation and Commands

Marks: 15

Objective

Demonstrate your understanding of the Linux filesystem and basic terminal commands.

Perform the following activities in your Linux environment.

Activity 1: Identify Your Environment

Execute appropriate commands to display:

1. Your current working directory.
2. Your logged-in username.
3. Your system hostname.
4. The Linux kernel version.
5. The current date and time.

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Activity 2: Explore the Filesystem

Navigate through the Linux filesystem and investigate the purpose of these directories:

- `/`
- `/home`
- `/etc`
- `/var`
- `/tmp`
- `/usr`
- `/bin`
- `/root`

For each directory, explain its primary purpose and give one example of the type of file or resource that might be found there.

Activity 3: Work with Files and Directories

Inside your home directory, create the following structure:

```
October_Assessment/
├── Networking/
│   ├── subnetting.txt
│   └── network_notes.txt
├── Linux/
│   ├── commands.txt
│   └── filesystem.txt
└── Reports/
    └── assessment_summary.txt
```

Use Linux terminal commands to:

1. Create the directories.
2. Create the empty files.
3. Display the directory structure.
4. List files, including hidden files.
5. Display the absolute path of your assessment directory.
6. Navigate between directories using both absolute and relative paths.

Activity 4: File Operations

Perform these operations within your assessment directory:

1. Write a short paragraph into `commands.txt`.
2. Display its contents.
3. Append another line without deleting the original content.
4. Copy `commands.txt` to a backup file.
5. Rename `filesystem.txt` to `linux_filesystem.txt`.
6. Move a copy of `subnetting.txt` into the `Reports` directory.

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7. Remove a temporary file that you created for this exercise.

Explain which command you used for each operation.

Evidence Required

- Screenshots of commands and outputs.
- Final directory structure.
- Contents of at least two files.
- A brief explanation of absolute paths versus relative paths.

Task 5 — Linux Permissions and File Management

Marks: 20

Scenario

You are a junior system administrator responsible for preparing files for a small IT team.

The team needs to store internal network documentation in a Linux directory. Some files should be readable by everyone on the team, while sensitive reports must have restricted access.

You must demonstrate your understanding of Linux file permissions and ownership.

Practical Activities

Activity 1: Understand Permissions

Create a file named `network_report.txt`.

Display its permissions and explain the output.

Answer:

1. What do `r`, `w`, and `x` represent?
2. What is the difference between file permissions and directory permissions?
3. What do owner, group, and others mean?
4. What does the permission string `rw-r--r--` mean?
5. How does the permission string differ for a directory?

Activity 2: Modify Permissions

Using your own test files:

1. Set `network_report.txt` to permission `640`.
2. Display the updated permissions.
3. Explain what `640` means in symbolic terms.
4. Change the file to permission `600`.
5. Explain how the change affects access by the group and other users.

Activity 3: Ownership

Investigate the current owner and group of your file.

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Answer:

1. Which command displays file ownership?
2. Which command changes file ownership?
3. Why might changing ownership require administrative privileges?
4. Why should sensitive reports not be accessible to every local user?

Do not change ownership of system files. Perform ownership exercises only on files you created and only where the necessary privileges are available.

Activity 4: Search and Inspect Files

Using files in your assessment directory:

1. Find all `.txt` files.
2. Search for a specific word inside a text file.
3. Display the first few lines of a file.
4. Display the last few lines of a file.
5. Count the number of lines, words, and characters in a file.
6. Identify the file type of a selected file.

Possible commands include:

- `find`
- `grep`
- `head`
- `tail`
- `wc`
- `file`
- `ls -l`
- `chmod`
- `chown`

Questions

1. What is the difference between `chmod` and `chown`?
2. Why is permission `777` generally a poor choice for sensitive files?
3. What is the difference between `>` and `>>` when redirecting command output?
4. What is the difference between standard output and standard error?
5. Why is least privilege important on a Linux system?

Evidence Required

- Before-and-after permission output.
- Explanation of numeric permissions.

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- Search results from your assessment directory.
- Screenshots showing successful file operations.

Task 6 — Linux Mini Challenge: Investigate a Small IT Environment

Marks: 15

Scenario

You have joined an IT support team as a junior Linux administrator.

A colleague reports that several files in a project directory are missing, a report cannot be read by another team member, and a temporary log file contains an error message that needs investigation.

Your task is to build a small simulated environment and investigate the issues using Linux commands.

Step 1: Create the Lab Environment

Inside `October_Assessment`, create the following:

```
Linux_Investigation/
├── logs/
│   └── application.log
├── reports/
│   └── network_report.txt
├── backup/
└── notes.txt
```

Populate `application.log` with these synthetic entries:

```
2026-10-16 09:10:12 INFO Service started
2026-10-16 09:11:03 INFO Configuration loaded
2026-10-16 09:12:45 ERROR Failed to connect to database
2026-10-16 09:13:10 WARNING Retrying connection
2026-10-16 09:14:02 ERROR Database connection timeout
2026-10-16 09:15:30 INFO Service stopped
```

Add a short paragraph to `network_report.txt`.

Step 2: Investigate the Files

Complete the following:

1. Display the full contents of `application.log`.
2. Search for all lines containing `ERROR`.
3. Count the number of error entries.
4. Display only the last three lines of the log.
5. Find all files inside `Linux_Investigation`.
6. Create a backup copy of the application log in the `backup` directory.

Step 3: Investigate Permissions

1. Display the current permissions of `network_report.txt`.

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2. Change the file permissions to `600`.
3. Verify the new permissions.
4. Explain who can read and modify the file under this configuration.

Step 4: Document Your Investigation

Create a file named `investigation_summary.txt` containing:

- The number of error entries found.
- The error messages identified.
- The location of the backup file.
- The permissions applied to the report.
- A short explanation of the commands used.

Questions

1. Which command did you use to search for errors?
2. How would you count the matching error lines?
3. Why is keeping a backup of an important log file useful?
4. What does permission `600` allow?
5. If a file is present but cannot be opened, what permissions and ownership details would you inspect?
6. How would you distinguish a missing file from a file that exists but cannot be accessed?

Evidence Required

- Final directory structure.
- Output showing the matching error lines.
- Error count.
- Backup file listing.
- Before-and-after permissions.
- Contents of `investigation_summary.txt`.

Important: Use synthetic log data and your own assessment directory. Do not modify system logs or unrelated files.

FINAL SUBMISSION GUIDELINES

Submit one PDF containing the completed networking and Linux assessment.

Required Sections

1. Cover Page
2. Task 1 — Office Network Design Case Study
3. Task 2 — Networking Concepts and Troubleshooting

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4. Task 3 — Practical Networking Commands
5. Task 4 — Linux Filesystem and Commands
6. Task 5 — Linux Permissions and File Management
7. Task 6 — Linux Investigation Challenge
8. Conclusion

Submission Checklist

- All six tasks completed.
- Subnetting calculations shown clearly.
- IP allocation table completed.
- Network diagram included.
- Networking questions answered.
- Linux commands executed successfully.
- Required directory structure created.
- File permissions demonstrated.
- Synthetic log investigation completed.
- Screenshots and command outputs included.
- Answers written in your own words.
- Final PDF reviewed before submission.

File Naming Convention

`OctoberBatch_StudentName_Networking_Linux_Assessment.pdf`

MARKING SCHEME

Task	Assessment Area	Marks
1	Office networking case study and subnetting	25
2	Networking concepts and troubleshooting	15
3	Practical networking commands	10
4	Linux filesystem and commands	15
5	Linux permissions and file management	20
6	Linux investigation mini challenge	15
Total		**100**

A Final Word

This assessment is designed to test your practical understanding, not just your ability to memorize commands.

For every task, explain what you did, why you used a particular command or configuration, and what you learned from the result.

Understand the network. Understand the system. Explain your decisions.

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